1) Connecting to MYSql Database and create a database if not exist.

```
import java.sql.*;
import java.util.Scanner;

public class CreateDatabase {

    private static final String JDBC_URL = "jdbc:mysql://localhost:3306/";
    private static final String USER = "root";
    private static final String PASSWORD = "";

    public static void main(String[] args) {
        Connection con = null;
        Statement statement = null;
        ResultSet resultSet = null;

        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter the name of the database to check/create: ");
        String databaseName = scanner.nextLine();

        try {
            // Register JDBC driver
            Class.forName("com.mysql.cj.jdbc.Driver");

            // Open a connection
            System.out.println("Connecting to MySQL server...");
            con = DriverManager.getConnection(JDBC_URL, USER, PASSWORD);

            // Create a statement
            statement = con.createStatement();

            // Check if the database exists
            resultSet = statement.executeQuery("SHOW DATABASES LIKE '" + databaseName + "'");
            if (resultSet.next()) {
                System.out.println("Database '" + databaseName + "' already exists.");
            } else {
                // Execute a query to create the database
                System.out.println("Creating database...");
                String sql = "CREATE DATABASE " + databaseName;
                statement.executeUpdate(sql);
                System.out.println("Database '" + databaseName + "' created successfully...");
            }
        } catch (SQLException se) {
            // Handle errors for JDBC
```

```
        se.printStackTrace();
      } catch (ClassNotFoundException e) {
        // Handle errors for Class.forName
        e.printStackTrace();
      } finally {
        // Clean-up environment
        try {
          if (resultSet != null) resultSet.close();
          if (statement != null) statement.close();
          if (con != null) con.close();
        } catch (SQLException se) {
          se.printStackTrace();
        }
        scanner.close();
      }
      System.out.println("Goodbye!");
    }
}
```

ii) Menu-Driven Program to create a database

```
import java.sql.Connection;

import java.util.Scanner;


public class MainApp {


    public static void main(String[] args) {

      // Open connection to the MySQL server

      Connection connection = MySqlConnection.openConnection();


      if (connection != null) {

        Scanner scanner = new Scanner(System.in);


        System.out.println("Do you want to create a new database or use an existing one?");

        System.out.println("1. Create new database");
```

```
        System.out.println("2. Use existing database");
        System.out.print("Enter your choice (1 or 2): ");
        int choice = scanner.nextInt();
        scanner.nextLine();  // Consume newline

        String databaseName;
        if (choice == 1) {
          System.out.print("Enter the name of the new database: ");
          databaseName = scanner.nextLine();
          CreateDatabase.createDatabase(connection, databaseName);
        } else if (choice == 2) {
          System.out.print("Enter the name of the existing database: ");
          databaseName = scanner.nextLine();
          System.out.println("You chose to use the existing database: " + databaseName);
        } else {
          System.out.println("Invalid choice. Exiting.");
          MySqlConnection.closeConnection();
          scanner.close();
          return;
        }

        // Close the connection
        MySqlConnection.closeConnection();
        scanner.close();
      }
    }
}
```